Do Now · retrieval

Lesson 085 · Searching, Sorting and Pathfinding Comparison

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the sorted-input rule for binary search?

Q2

State merge sort's worst-time complexity.

Q3

State f score in A*.

Q4

Push A,B,C onto a stack, Pop twice, then Push D. Give bottom-to-top order.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Start from the problem and input: searching, ordering and weighted paths are different problem classes.

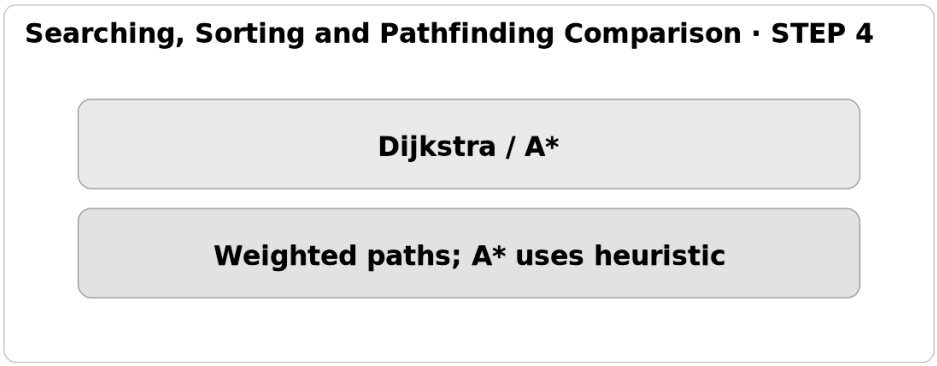
Core idea 2

Apply — A fastest method under one set of assumptions may be invalid or less suitable on another dataset; include preprocessing and memory costs.

Core idea 3

Evaluate — Choose using correctness, worst-case growth, memory, stability where relevant, and an independent trace on new values.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

suitability — How well method fits input/task.

precondition — Requirement for correct use.

complexity — Growth of time/space.

heuristic — Estimate guiding search.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

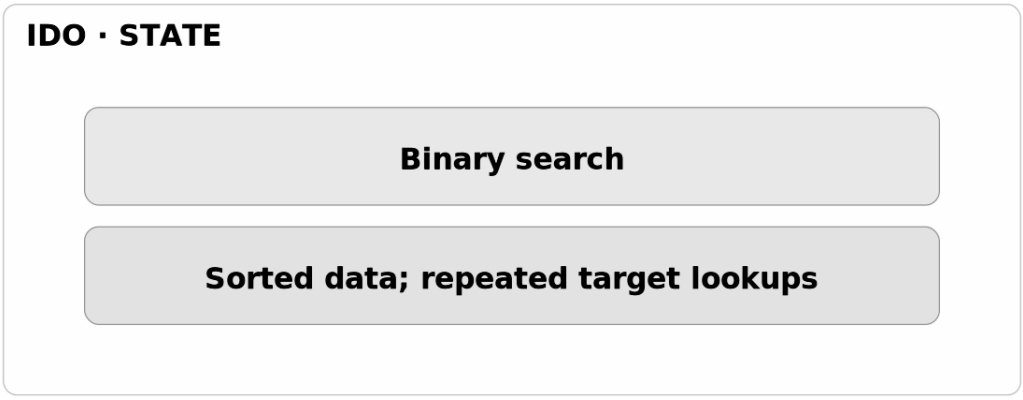
Problem

Choose algorithm for 200 sorted pupils by ID with repeated lookups.

Follow the worked method

1. Method and result — Binary search O(log n) is valid because identifiers are sorted; if indexed dictionary is available, keyed lookup is another option.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Method and result
Dijkstra finds least-cost path with non-negative edge weights; do not confuse with array search.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Identify suitable algorithm for (a) unsorted match (b) sorted match (c) non-negative weighted shortest path. [6]

Question 2

Compare bubble, insertion, merge and quick sort worst-case time and typical auxiliary memory. [8]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A college searches a million unsorted attendance records once and later queries sorted year-group reports repeatedly. Evaluate algorithms/data processing choices. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

All standard algorithms can be swapped freely without checking their input assumptions.

Use key vocabulary from memory

1. Explain suitability in your own words.

2. Explain precondition in your own words.

3. Explain complexity in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.